BIOL6130 Final Project: data preparation

Read the data

```
library(sf)
library(terra)
library(dplyr)
library(ggplot2)
library(tidyterra)

photo_2018 <- rast("Raw Data/dgt_ortho2018_context.tif")
photo_2025 <- rast("Raw Data/dgt_ortho2025_context.tif")
land_cover <- read_sf("Raw Data/cos2018_land_cover.geojson")
forest_2018 <- read_sf("Raw Data/cos2018_eucalyptus.geojson")
forest_2025 <- read_sf("Raw Data/cos2025_eucalyptus.geojson")
road <- read_sf("Raw Data/cos2018_roads.geojson") %>%
  filter(ogc_fid == 147070)

# Crop climate data to the current B window.
area <- as.polygons(ext(photo_2018), crs = crs(photo_2018))
window <- project(area, "EPSG:4326")
climate_file <- "Raw Data/PRT_wc2.1_30s_bio.tif"
climate_url <- paste0(
  "https://geodata.ucdavis.edu/climate/worldclim/2_1/tiles/iso/",
  "PRT_wc2.1_30s_bio.tif")
if (!file.exists(climate_file)) {
  download.file(climate_url, climate_file, mode = "wb", quiet = TRUE)
}
climate <- rast(climate_file)[[c(1, 12)]] %>%
  crop(window, snap = "out")
names(climate) <- c("temperature", "precipitation")
```

Inspect the contents

```
glimpse(land_cover, width = 75)
```

```
Rows: 6,063
Columns: 12
$ ogc_fid      <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367, 1368~
$ cos18v4_n4_c <chr> "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1~
$ cos18v4_n4_l <chr> "Áreas edificadas residenciais contínuas predominant~
$ id           <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367, 1368~
$ cos18v4_n3_c <chr> "111", "111", "111", "111", "111", "111", "111", "11~
$ cos18v4_n3_l <chr> "Áreas edificadas residenciais contínuas", "Áreas ed~
$ cos18v4_n2_c <chr> "11", "11", "11", "11", "11", "11", "11", "11", "11"~
$ cos18v4_n2_l <chr> "Áreas edificadas residenciais", "Áreas edificadas r~
$ cos18v4_n1_c <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1~
$ cos18v4_n1_l <chr> "Territórios artificializados", "Territórios artific~
$ area_ha      <dbl> 2.5112068, 4.9795026, 1.3266062, 7.0387354, 2.877182~
$ geometry     <MULTIPOLYGON [°]> MULTIPOLYGON (((-8.251433 4..., MULTIPO~
```

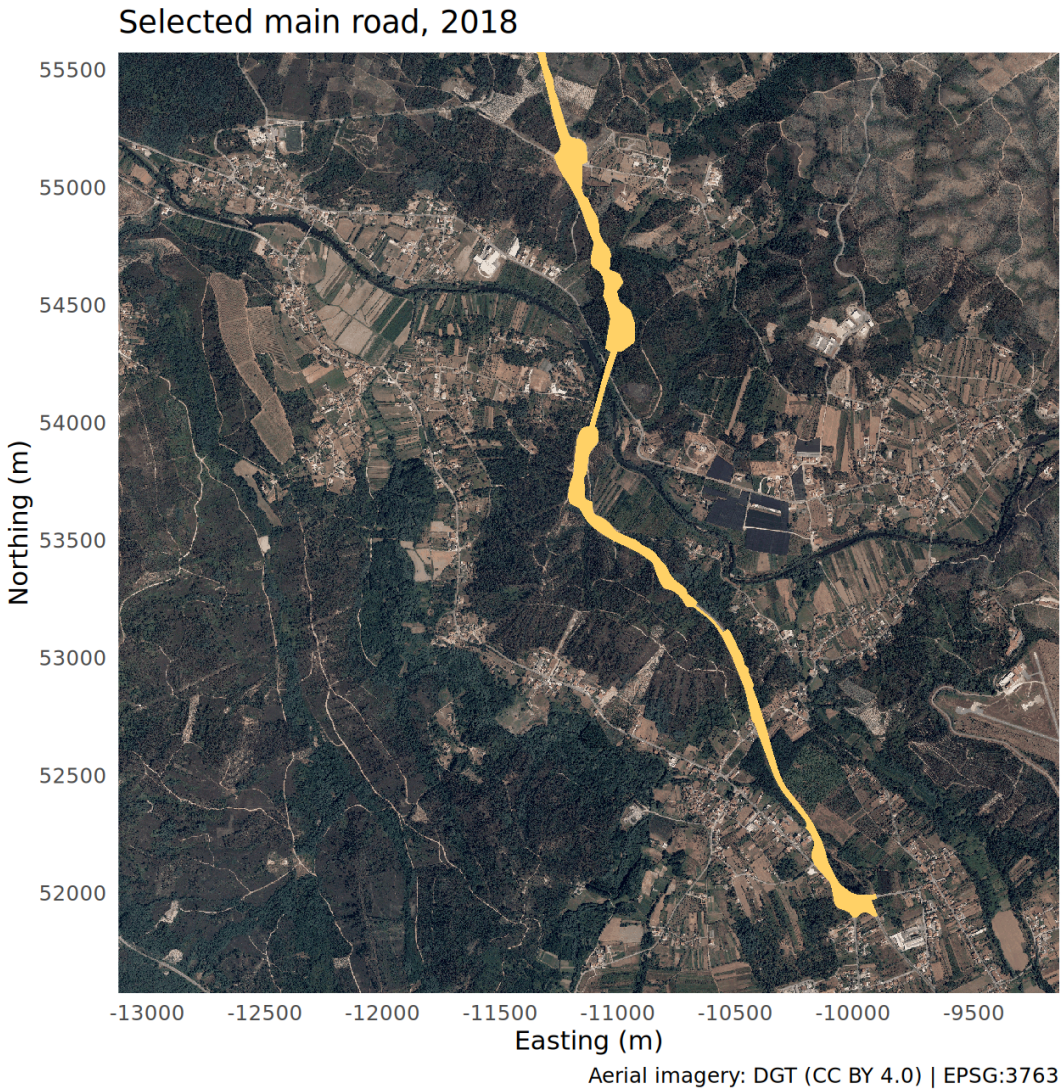
Draw the maps

B is the current preview window. The focal road is COS ID 147070; side roads are deferred. Sampling will focus on land not mapped as eucalyptus in 2018. The final sampling extent and distance groups remain undecided.

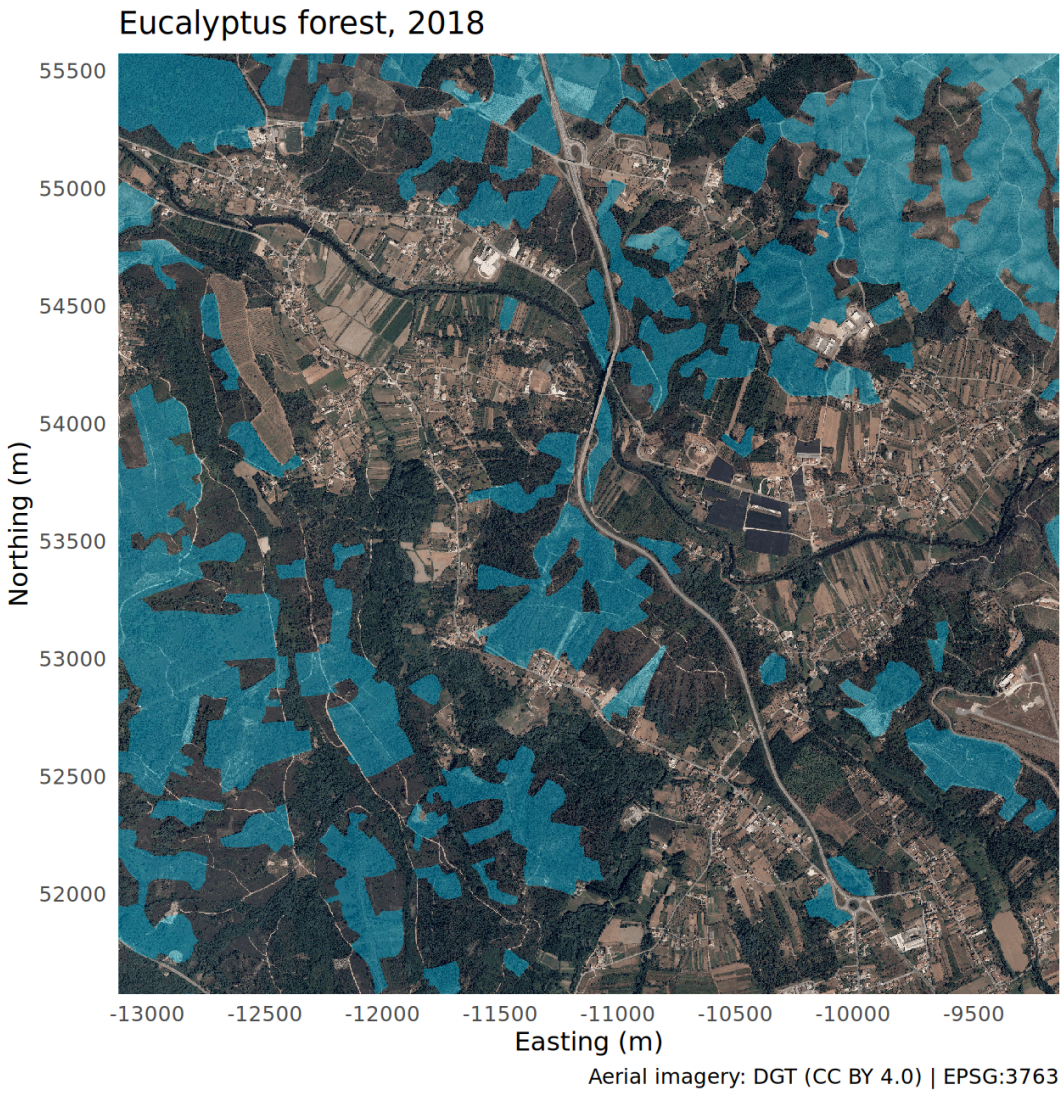
The climate maps show 1970–2000 averages at 30-arc-second resolution (about 710 by 925 m here). The 2025 aerial image provides geographic context.

```
# tidyterra draws the photograph; ggplot2 adds the data layer and map layout.
draw_map <- function(photo, layer, title) {
  ggplot() +
    geom_spatraster_rgb(data = photo, interpolate = FALSE,
                        maxcell = ncell(photo)) +
    layer +
    coord_sf(crs = 3763, datum = st_crs(3763), expand = FALSE,
             xlim = c(xmin(photo), xmax(photo)),
             ylim = c(ymin(photo), ymax(photo))) +
    labs(title = title, x = "Easting (m)", y = "Northing (m)",
         caption = "Aerial imagery: DGT (CC BY 4.0) | EPSG:3763") +
    theme_minimal()
}
```

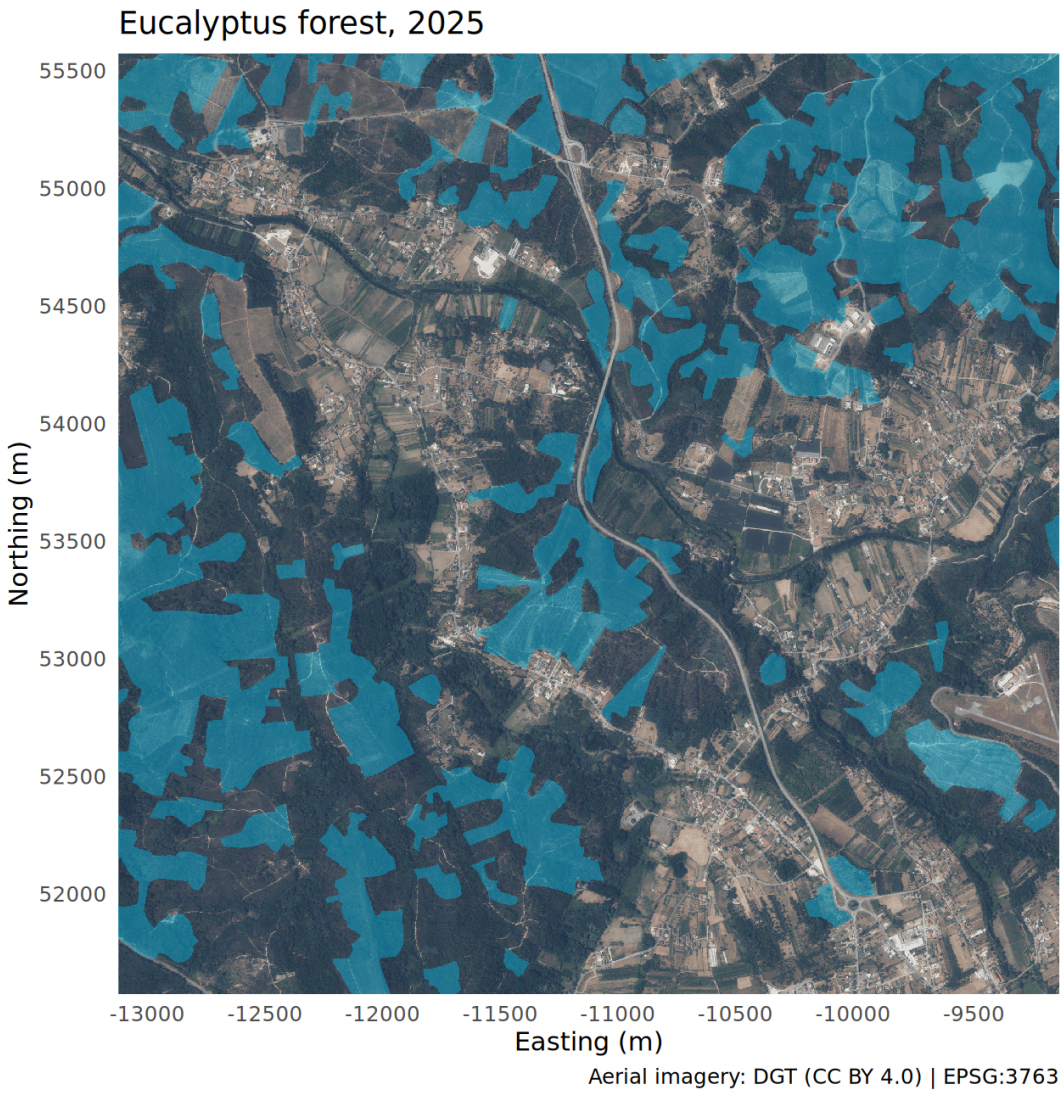
```
draw_map(photo_2018,
         geom_sf(data = road, fill = "#FFD166", colour = NA),
         "Selected main road, 2018")
```



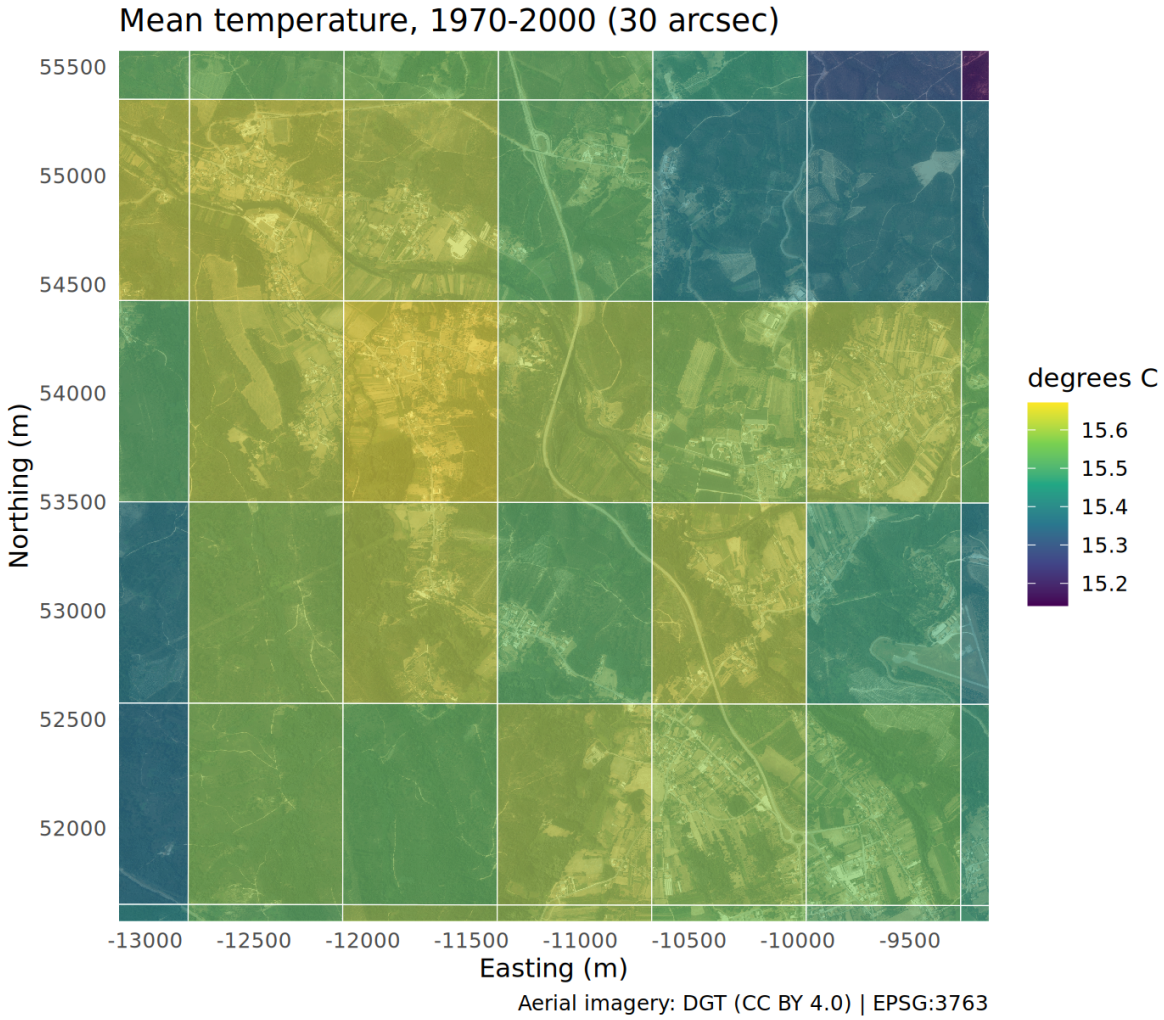

```
draw_map(photo_2018,
         geom_sf(data = forest_2018, fill = "#00BFE8",
                 colour = NA, alpha = 0.4),
         "Eucalyptus forest, 2018")
```



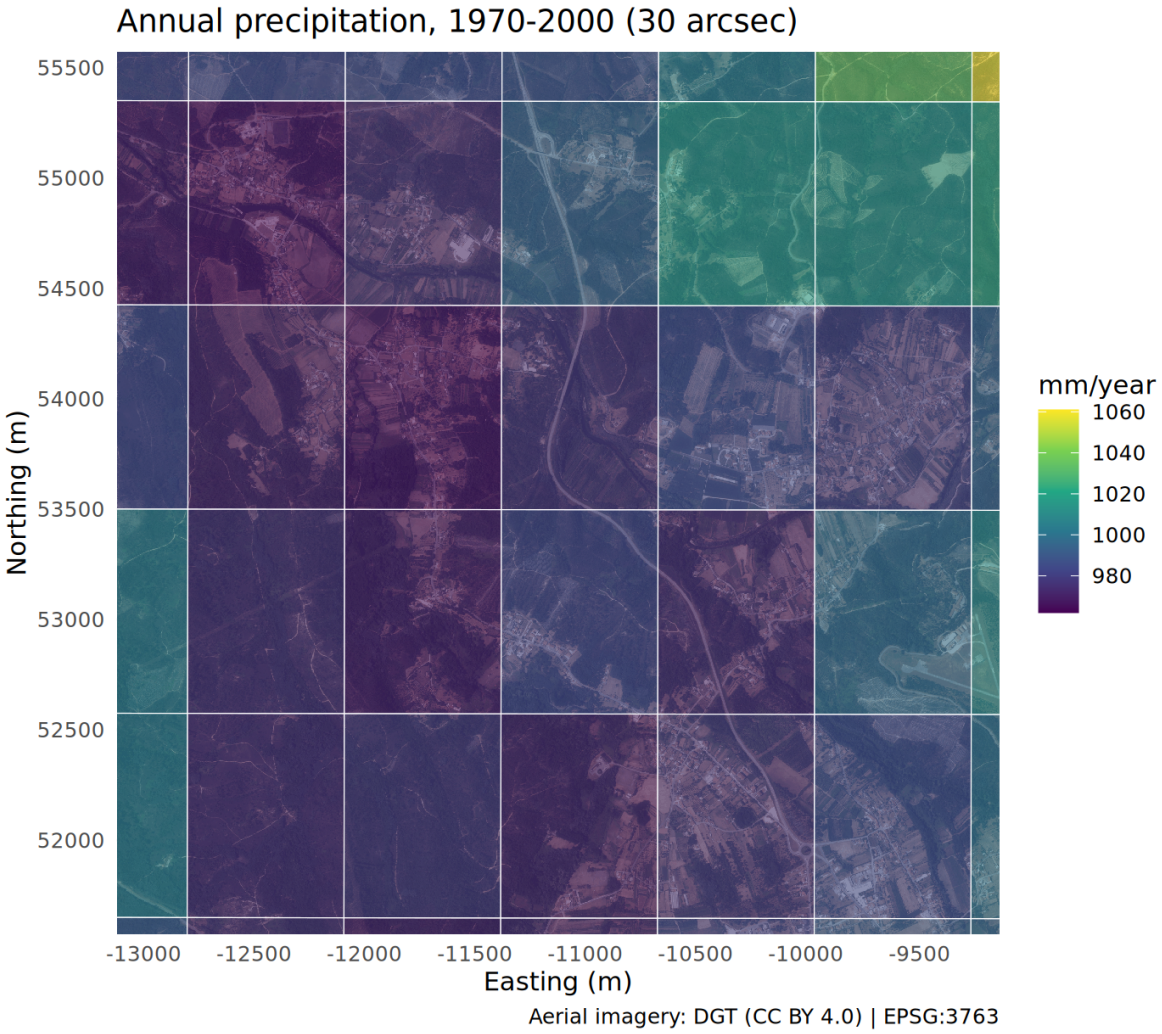

```
draw_map(photo_2025,
         geom_sf(data = forest_2025, fill = "#00BFE8",
                 colour = NA, alpha = 0.4),
         "Eucalyptus forest, 2025")
```




```
# Keep the original climate cells and values.
cells <- as.polygons(climate, aggregate = FALSE, round = FALSE) %>%
  st_as_sf()
draw_map(photo_2025,
         geom_sf(data = cells, aes(fill = temperature),
                 colour = "white", alpha = 0.55),
         "Mean temperature, 1970-2000 (30 arcsec)") +
  scale_fill_viridis_c(name = "degrees C")
```




```
draw_map(photo_2025,
         geom_sf(data = cells, aes(fill = precipitation),
                 colour = "white", alpha = 0.55),
         "Annual precipitation, 1970-2000 (30 arcsec)") +
  scale_fill_viridis_c(name = "mm/year")
```



Sources and use

- Direcao-Geral do Territorio (DGT): COS2018v4 and COS2025v1, CC BY 4.0. The 2018 inputs were downloaded on 2 November 2026; the 2025 input is the verified 28 October 2026 snapshot.
- DGT orthophotography: 2018 and 2025 true-colour aerial imagery, CC BY 4.0. The original product has 25 cm pixels; these WMS background exports use a 2 m display grid for area B and were downloaded on 4 November 2026.
- Copernicus GLO-30: surface elevation, under the Copernicus DEM licence. © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018 provided under

COPERNICUS by the European Union and ESA; all rights reserved.

- WorldClim 2.1: BIO1 and BIO12 from the Portugal file, 1970–2000, 30-arc-second resolution; Fick and Hijmans (2017), doi:10.1002/joc.5086. The TIFF is downloaded from WorldClim when absent and is excluded from Git because redistribution requires permission.

Map functions: tidyterra and ggplot2.